

Temporal Causal Discovery and Generative Prediction of Vehicular CO₂ emission

Supplementary Material

CCS Concepts

• **Computing methodologies** → Causal reasoning and diagnostics; Temporal reasoning; Probabilistic reasoning; Neural networks; • **Mathematics of computing** → Causal networks.

Keywords

Causal discovery, Causal inference, Time series forecasting, Variational Recurrent Neural Networks, Vehicular emission forecasting

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Contents

We have arranged the additional information and analysis in this supplementary material in different sections and subsections. The arrangement of sections and their contents are as follows:

- **§1 Synthetic Data Generation and Splitting:** Consists of all necessary equations used for synthesizing the Hénon chaotic maps [11] and Lorenz-96 [12] data.
- **§2 Details on OBD-II dataset:** Necessary details and feature selection of OBD-II dataset.
- **§3 Design Principles:** Details and novel considerations of our proposed causal recurrent variational autoencoder architecture.
- **§4 Evaluation Metrics:** Details on the evaluation metrics used for quantitative comparison with the baselines.
- **§5 Results on Causal Discovery:** Additional results on causal discovery on Hénon and Lorenz-96 datasets.
- **§6 Additional Baseline Comparisons:** Comparison with additional relevant baseline methods from past years.
- **§7 Implementation setup:** Implementation details such as device and model configurations.

Source code : The source code can be found here for review - <https://anonymous.4open.science/r/causal-obd-co2-0A0C>.

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1 Synthetic Data Generation and Splitting

Following these equations, we have generated the Hénon chaotic maps,

$$x_{t+1}^1 = 1.4 - (x_t^1)^2 + 0.3x_{t-1}^1, \quad (1)$$

$$x_{t+1}^d = 1.4 - (ex_t^{d-1} + (1-e)x_t^d)^2 + 0.3x_{t-1}^d, \quad (2)$$

where $d \in [2, M]$, M is the dimensionality, $e = 0.3$ and $d = 10$. The true causal relations $x^d \rightarrow x^{d+1}$ and self-causal relations $x^d \rightarrow x^d$ in the corresponding adjacency matrix should be 1. The lag length has been taken as 2.

Similarly, we have used the following equations for simulating the M-dimensional Lorenz-96 chaotic maps,

$$\frac{dx_t^p}{dt} = (x_t^{p+1} - x_t^{p-2})x_t^{p-1} - x_t^p + F \quad (3)$$

where $x_t^{-1} = x_t^{M-1}$, $x_t^0 = x_t^M$, $x_t^{M+1} = x_t^1$, p is the index variable and F is the forcing constant that dictates the degree of nonlinearity and chaos in the series.

We generated around 5000 samples for each dataset. For the Hénon and Lorenz-96 systems, we sample initial values from a typical Gaussian distribution and then infer the trajectories via transition functions.

The generated data is then divided into several chunks of consecutive timesteps. Each of the chunks is considered as a single instance. We divide the set of instances into train and validation sets in a ratio of 80:20. For all the experiments, we have taken the chunk size $X_{1:T} = 30$. As shown in Figure 2 in the main manuscript, each chunk is then divided into two parts, $X_{1:T-1}$ and $X_{T:T-1}$ where $\tau = 15$.

2 Details on OBD-II dataset

The OBD-II dataset was introduced in the work by [16]. Mainly, the authors utilized an OBD Bluetooth adapter and a smartphone to collect data from two vehicles, then analyzed the correlation between RPM and speed data to ascertain whether it indicates the vehicle's present gear. The two cars belong to the same size category, although their manufacturers and engine power differ. The OBD-II interface, extensively utilized, was developed to standardize the physical connection, its pin configuration, signaling protocols, and message format. It is utilized for aftermarket maintenance, granting access to engine trouble codes and describing mechanics about breakdowns across the vehicle, therefore conserving significant time in diagnosing the source of issues.

The dataset contains around 43 features, including the car and driver identifiers, features directly obtained from the OBD scanner, calculated features, and variables obtained by smartphone sensors such as GPS, Altitude and Air pressure. As our work focuses on vehicle sensor data, we have manually removed the features other than those directly obtained from the OBD sensors. We leave the investigation of causal effects of other variables in the dataset in CO₂

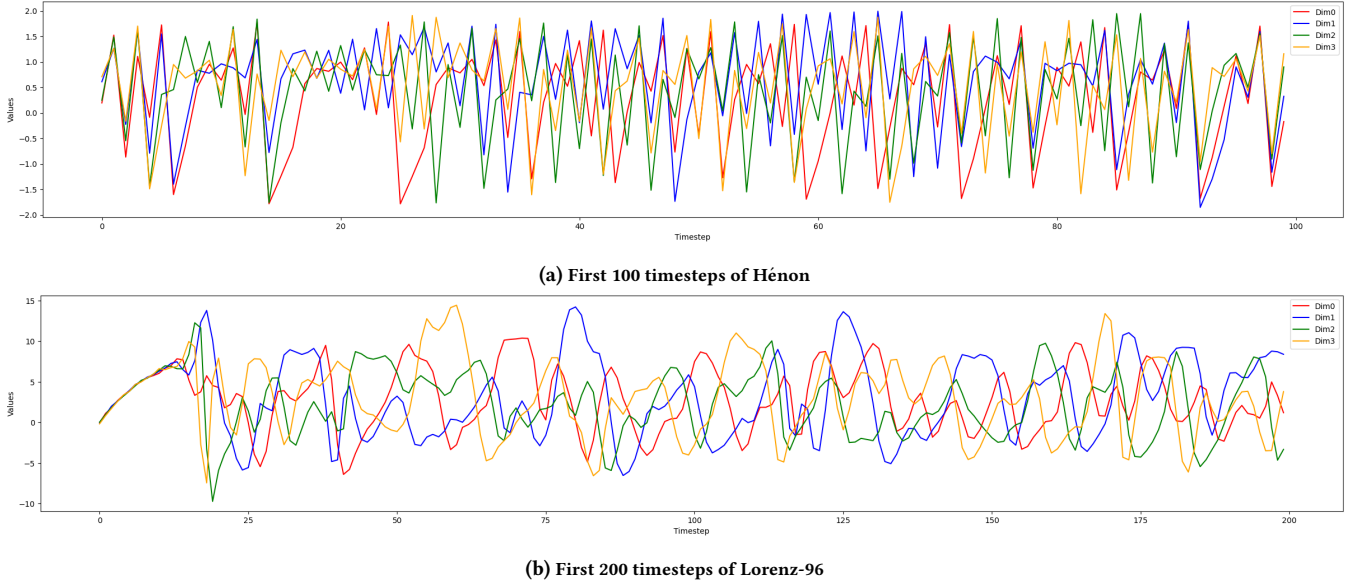


Figure 1: Visualization of the first 4 dimensions of the generated Hénon and Lorenz-96 datasets

emission as a future work. The selected 15 OBD features, excluding the car identifier (Car ID), used in this work are summarized in Table 1.

The total number of instances in the dataset is 91,794 in which 81,095 instances come from vehicle 1 (Car ID = 1) and the rest of the 6,699 instances come from vehicle 2 (Car ID = 2). In our experiment, we have used the instances of vehicle 1 exclusively as the training set and that of vehicle 2 as the validation set.

3 Design principles

There are a few unique designs that distinguish our proposed framework from the conventional recurrent variational autoencoder. These include the multi-head decoder, the unidirectional inputs, and the error compensation module.

3.1 Unidirectional inputs

The original VRAE, as well as its other notable variations [4, 7], employs the input of $x_{T-\tau:T-1}$ for both the encoder and the decoder. Before decoding, the information that pertains to the entire sequence is encoded in this manner. The estimation of $p(x_t|x_{1:t})$ is performed via these procedures, as opposed to $p(x_t|x_{1:t-1})$, which means that the future input values occurring at time t cannot be utilized in the conditional variable. The concept that [14] used to describe this phenomenon is known as causal conditioning. Due to the fact that it “peeps on the future,” it is impossible for it to ever identify causality in the way that Granger [8] defines it. This is because it violates the fundamental rules of Granger causality within the context of causal discovery.

For the sake of providing evidence for our argument, we performed two separate experiments using a synthetic 10-dimensional linear autoregressive process with lag 3. Mathematically it can be

written as,

$$x_t = \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \alpha_3 x_{t-3} + \epsilon_t \quad (4)$$

where $\epsilon_t \sim \mathcal{N}(0, 1)$. The true causal matrix for the synthetic process can be obtained by all non-zero elements of $\alpha_1 + \alpha_2 + \alpha_3$.

We choose to use the input of the encoder $x_{T-\tau:T-1}$ rather than $x_{T-2\tau-1:T-1}$. As can be seen in Figure 2, our model is able to identify the majority of true causal relationships. On the other hand, the altered baseline, which has an encoder that looks at future values, is unable to identify the causal directions that exist between the majority of time series pairings.

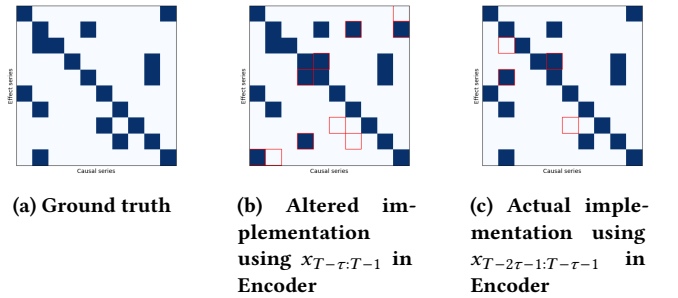


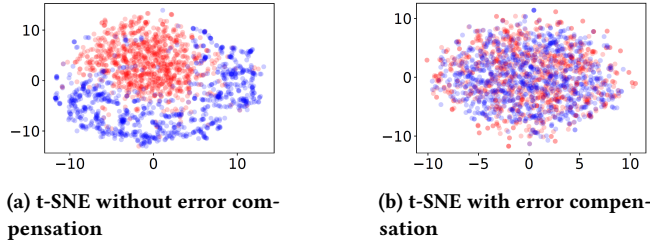
Figure 2: Visual comparison of causal graph adjacency matrices recovered by altered and actual implementations of our model with the ground truth causal matrix. Wrongly identified causal relations have been highlighted using red rectangles.

3.2 Error Compensation Module

We subsequently validate the necessity of the error-compensation module. We analyze the time series generation outcomes of the

Table 1: Summarization of the OBD features used in our study

Feature name	Description
Intake Air Temperature	Temperature of the air utilized in the air-fuel combination.
Engine Temperature	Current temperature of the engine coolant fluid.
Engine load	Degree of stress or force exerted on an engine during operation.
Engine RPM	Engine's Revolutions per Minute.
Fuel flow	Instantaneous usage of fuel by the engine.
Fuel level	Current amount of fuel.
Instant Mileage	Immediate fuel use per kilometer.
Average Mileage	Average fuel use per kilometer for each log.
Speed	Speed reading from odometer.
Ambient Temperature	Temperature of air around the vehicle.
Air pedal	Percentage, indicating how fully the pedal is pressed.
Acceleration	Change of speed between two observations.
Air drag force	Measure of aerodynamic resistance that opposes a vehicle's motion.
Instant CO ₂ emission	Instantaneous CO ₂ emission reading.
Average CO ₂ emission	Average CO ₂ emission reading.

**Figure 3: Visual comparison using t-SNE plots of the linear autoregressive process. Red samples represent the original time series, and blue samples represent the predicted time series.**

original model and its inferior variant lacking error correction. We employ t-SNE [19] to display the produced samples. An effective generative model should produce a distribution that closely resembles the real data distribution. Figure 3 illustrates that the error-compensation network results in a substantial increase in performance. Samples produced by the model without error correction rapidly converge to values approaching zero. We adjusted the parameters $\{\alpha_1, \alpha_2, \alpha_3\}$ to prevent divergence, and the actual $x_t = \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \alpha_3 x_{t-3}$ did converge. The model without error compensation encapsulates the dynamics $p(x_t | x_{1:t-1})$, while disregarding ϵ_t .

The description and mathematical equations of the multi-head decoder have been given in the main manuscript.

4 Evaluation Metrics

Here we have provided a brief description and equations of the quantitative evaluation metrics we have used in our work.

Area under Receiver Operating Characteristic curve (AUROC). The Receiver Operating Characteristic (ROC) curve is a graphical representation that plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at different threshold values. The AUROC represents the area under this curve, providing a single

scalar value that summarizes the model's performance. The AUC ranges from 0 to 1, where, an AUC of 0.5 indicates no discriminative ability and an AUC of 1.0 indicates perfect discrimination between classes. It can be mathematically represented as shown in Equation 5.

$$\text{AUROC} = \sum_{i=1}^N (\text{FPR}_i - \text{FPR}_{i-1}) \cdot \text{TPR} \quad (5)$$

where TPR and FPR stand for True Positive Rate and False Positive Rate respectively and they can be given as follows,

$$\text{TPR} = \frac{\# \text{True Positives}}{\# \text{True Positives} + \# \text{False Negatives}}$$

$$\text{FPR} = \frac{\# \text{False Positives}}{\# \text{False Positives} + \# \text{True Negatives}}$$

We use the AUROC score as a quantitative indicator while assessing the calculated causal adjacency matrices by comparing them to the ground truth. The numerical results have been given in the main manuscript.

Maximum Mean Discrepancy (MMD). Maximum mean discrepancy (MMD) [9] is extensively employed to quantify the divergence between two distributions. It assesses whether two sample sets—one synthetic and the other derived from real data—originate from the same distribution. We have used this method to quantitatively evaluate the chosen baselines and our model and compare the results in the task of generative forecasting. Formally, it can be written as shown in Equation 6.

$$\text{MMD}(x, \hat{x}) = \frac{1}{n^2} \kappa(x_i, x_j) + \frac{1}{n^2} \kappa(\hat{x}_i, \hat{x}_j) - \frac{2}{n^2} \kappa(x_i, \hat{x}_j) \quad (6)$$

where x and $\hat{x}$ are the actual data and the predicted data, indexed by i and j , respectively; the kernel κ is designated as the Gaussian kernel. We compute the average results for a kernel size bandwidth of $[0.01; 0.1; 1; 10; 100]$, consistent with [6]. The MMD provides greater informational value than both generator and discriminator loss in GAN-based models [3], so we use early stopping to achieve the minimal MMD for generative models.

Root Mean Squared Error (RMSE). RMSE is one of the widely used metrics in forecasting models. It is defined as the square root

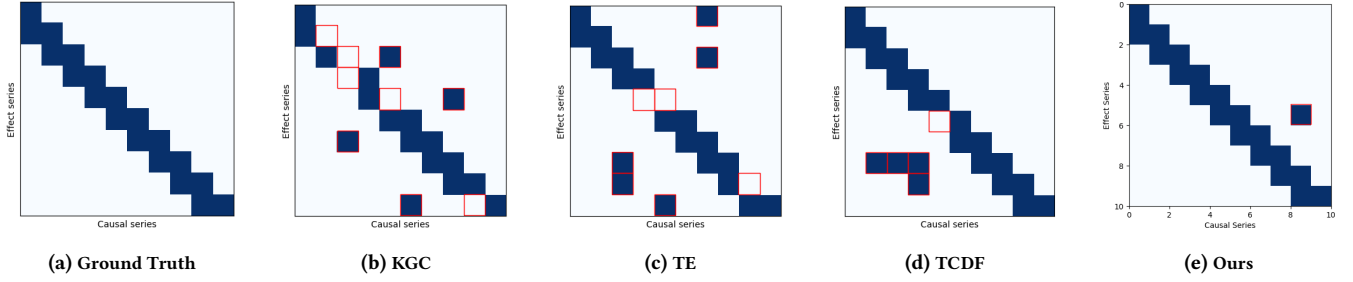


Figure 4: Estimated causal matrices from the additional baselines for causal discovery and our proposed approach on Hénon maps. Wrongly identified causal relations have been highlighted using red rectangles.

of the average of the squared differences between predicted and observed values. It essentially represents the standard deviation of the residuals, indicating how concentrated the data points are around the hyperplane of best fit. A lower RMSE value signifies a better fit, indicating that the model’s predictions are closer to the actual outcomes. Equation 7 shows the mathematical representation.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (7)$$

where y_i is the ground truth, $\hat{y}_i$ the predicted value, and N is the number of instances.

Mean Average Error (MAE). MAE is one of the widely used metrics in forecasting models. It is defined as the average of the absolute differences between predicted values and actual values. It quantifies how far off predictions are from the true values, providing a clear measure of prediction accuracy. The MAE is always non-negative, and a value of zero indicates perfect predictions. Lower MAE values suggest better model performance, while higher MAE values indicate larger discrepancies between predicted and actual values. MAE can be mathematically written as shown in Equation 8.

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (8)$$

where y_i is the ground truth, $\hat{y}_i$ is the predicted value and N is the number of instances.

5 Results on Causal Discovery

We have given the absolute adjacency matrices of Granger causal networks and match them with those calculated by our proposed approach and chosen baseline methodologies for causal discovery. In conventional Granger causality methods, we choose thresholds by optimizing AUROC and exclude any values below these thresholds. For neural network-based techniques, we may directly present the estimated matrices. As seen in Figures 5 and 6, our model surpasses the majority of baseline models and attains competitive outcomes among neural network-based approaches for both the Hénon and Lorenz-96 datasets. Red squares denote false positive or negative components.

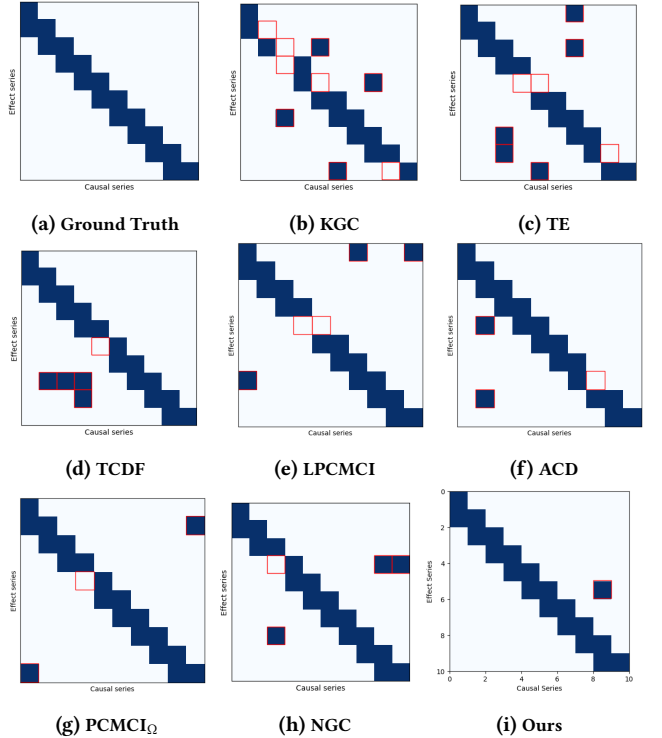


Figure 5: Estimated causal matrices from the chosen baselines for causal discovery and our proposed approach on Hénon maps. Wrongly identified causal relations have been highlighted using red rectangles.

6 Additional Baseline Comparisons

In addition to the comparison with the recent state-of-the-art methods in the main manuscript, we have chosen few other significant works closely related to our approach.

6.1 Causal Discovery

For the task of causal discovery, we have chosen four popular methods: Kernel Granger Causality (KGC) [13], which utilizes the kernel trick to extend Granger causality from linear to non-linear scenarios; Transfer Entropy (TE) [18], estimated using the matrix-based

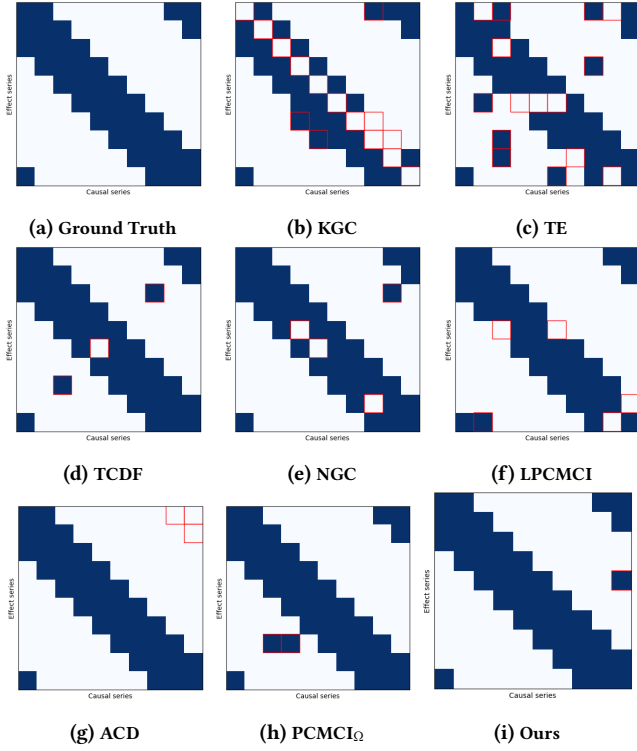


Figure 6: Estimated causal matrices from the chosen baselines for causal discovery and our proposed approach on Lorenz-96. Wrongly identified causal relations have been highlighted using red rectangles.

Rényi’s α -order entropy functional [5]; Temporal Causal Discovery Framework (TCDF) [15] incorporates an attention mechanism into a neural network. KGC and TE utilize information-theoretic measures, that is, focusing on independence or conditional independence, and employ postprocessing techniques such as hypothesis testing. On the other hand, TCDF and our proposed approach are neural network-based methods that actively and automatically extract causal relationships during the learning process.

The quantitative comparison has been done in terms of AUROC, which has been shown in the first part of Table 2. Figure 4 represents a qualitative comparison on Hénon maps in terms of the estimated causal adjacency matrix. It is to be noted that neural network-based methods significantly outperform classical methods KGC and TE. This is because conventional methods lack the ability to identify self-cause.

6.2 Generative Forecasting

For the generative forecasting task, we have chosen three baselines: Time-series generative adversarial network (TGAN) [20], a type of generative adversarial network that incorporates transition dynamics within the GAN framework, Variational RNN (VRNN) [2], and Variational recurrent autoencoder (VRAE) [10]. As these models are generative models, during inference, they generate data from a random noise sampled from a Gaussian distribution $\mathcal{N} \sim (0, 1)$ [1].

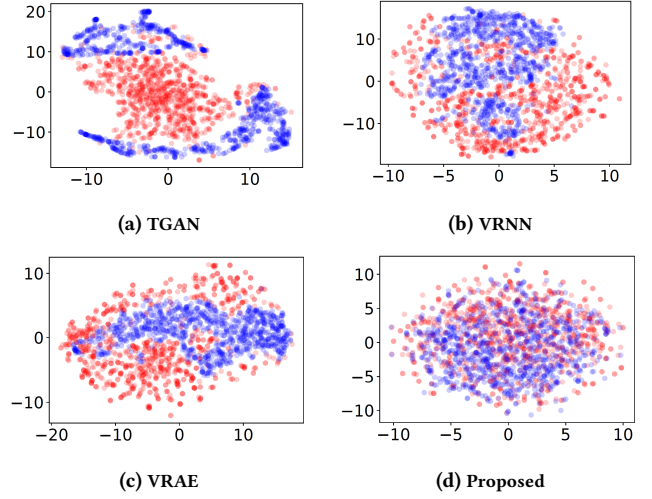


Figure 7: t-SNE visualizations on Hénon. Red samples represent actual time series $x_{t:T-1}$, and blue samples represent the forecasted time series $\hat{x}_{t+1:T}$.

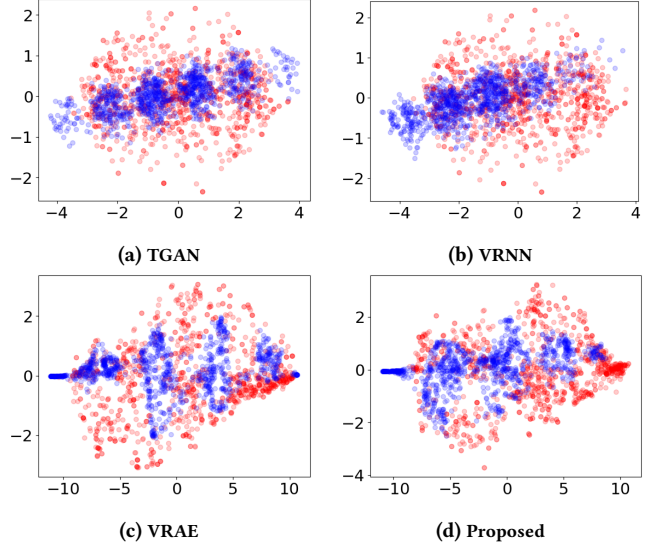


Figure 8: t-SNE visualizations on the OBD-II dataset. Red samples represent actual time series $x_{t:T-1}$, and blue samples represent the forecasted time series $\hat{x}_{t+1:T}$.

We assess the quality of the generated time series using t-SNE by projecting them into a 2-dimensional space. Figure 7 and Figure 8 show the qualitative comparisons with the above-mentioned methods on Hénon maps and the OBD-II dataset, respectively. The quantitative comparison has been done in terms of MMD and RMSE, which have been reported in the second part of Table 2.

7 Implementation setup

The whole system has been implemented in PyTorch v2.4.0 with support for CUDA v12.5 and trained on a single NVIDIA V100 GPU

Table 2: Quantitative comparison for causal discovery and forecasting. Best results have been given in bold. The second best results have been underlined.

Task	Metric	Model	Hénon	Lorenz-96	OBD-II
Causal Discovery	AUROC	KGC	0.462	0.635	NA
		TE	0.465	0.410	NA
		TCDF	0.905	0.870	NA
		Proposed	0.980	0.975	NA
Generative Forecasting	MMD	TGAN	0.475	0.039	2.109
		VRNN	0.321	0.043	1.915
		VRAE	<u>0.125</u>	0.011	<u>1.877</u>
		Proposed	0.119	<u>0.019</u>	1.839
	RMSE	TGAN	0.291	0.124	0.901
		VRNN	0.179	0.129	<u>0.618</u>
		VRAE	<u>0.125</u>	<u>0.123</u>	0.756
		Proposed	0.120	0.109	0.492

with 32GB GPU memory. We have chosen single-layer LSTM [17] for implementing the main causal recurrent model and the error compensation model. We have set the value of λ in PGD as 0.1 and kept the default values of β_1 and β_2 in Adam, which are 0.9 and 0.999, respectively. The maximum number of epochs is 2000, and the learning rate γ has been set at 0.03. Additionally, we have incorporated early stopping regularization based on the validation loss, which stops the training if the validation loss does not decrease for 200 consecutive epochs.

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